Literature Review: Suicide Risk Detection Using Natural Language Processing

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August 31, 2026

Abstract

This literature review examines existing research on suicide risk detection using natural language processing (NLP) and machine learning techniques. We survey approaches ranging from traditional text classification to modern large language model (LLM) based methods, with particular emphasis on cross-cultural and multilingual contexts. The review covers data collection methodologies, feature extraction techniques, clustering approaches, and risk scoring frameworks. We identify gaps in current research related to South Asian contexts and propose directions for future work in culturally-aware suicide risk detection systems.

Keywords: Literature review, Suicide risk detection, NLP, Machine learning, Cross-cultural analysis, Mental health

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1 Introduction

1.1 Background

Suicide remains a critical global public health challenge, with approximately 700,000 deaths annually worldwide [1]. The World Health Organization identifies suicide as a leading cause of death among young people aged 15–29 years. Understanding risk factors and developing early detection systems has become a priority for public health research.

1.2 Motivation

Traditional approaches to suicide risk assessment rely on clinical interviews and self-report questionnaires, which have limitations in scalability and early detection. The proliferation of digital media and news reporting has created new opportunities for automated risk detection through text analysis.

1.3 Scope

This review covers:

- NLP-based suicide risk detection methods
- Machine learning and deep learning approaches
- Cross-cultural and multilingual considerations
- Data sources and collection methodologies
- Ethical considerations in suicide research

2 Suicide Risk Detection Methods

2.1 Traditional Machine Learning Approaches

2.1.1 Feature-Based Classification

Early approaches relied on handcrafted features and traditional classifiers:

- **Bag-of-Words (BoW):** Simple frequency-based representations that capture term occurrence patterns [2]
- **TF-IDF:** Term Frequency-Inverse Document Frequency weighting that emphasizes discriminative terms [3]
- **Sentiment Analysis:** Lexicon-based and machine learning sentiment classifiers adapted for suicide-related text [4]

2.1.2 Classical Algorithms

Commonly applied algorithms include:

Table 1: Traditional ML Methods for Suicide Detection

Algorithm	Reference	Key Finding
SVM	Coppersmith et al. (2014)	High accuracy in social media posts
Random Forest	Cao et al. (2020)	Effective feature selection
Logistic Regression	De Choudhury et al. (2013)	Baseline performance
Naive Bayes	ri Nawaz et al. (2020)	Fast inference

2.2 Deep Learning Approaches

2.2.1 Neural Network Architectures

- **CNN-based:** Convolutional neural networks for local feature extraction in text classification [5]
- **RNN/LSTM:** Recurrent architectures for sequential text modeling [6]
- **Transformer-based:** Attention mechanisms enabling context-aware representations [7]

2.2.2 Pre-trained Language Models

Recent work has leveraged transfer learning with large pre-trained models:

- **BERT:** Bidirectional Encoder Representations from Transformers [8]
- **RoBERTa:** Robustly Optimized BERT Approach [9]
- **IndicBERT:** Multilingual model for Indian languages [10]
- **Llama-3:** Large language model with multilingual capabilities [11]

2.3 Large Language Model (LLM) Approaches

2.3.1 Zero-shot and Few-shot Learning

LLMs have demonstrated capability in suicide risk detection without task-specific training:

- Prompt engineering for risk classification
- In-context learning with minimal examples
- Chain-of-thought reasoning for explainable detection

2.3.2 Fine-tuning Strategies

- Task-specific fine-tuning on suicide-related datasets
- Domain adaptation for mental health language
- Parameter-efficient fine-tuning (LoRA, QLoRA)

3 Data Sources and Collection

3.1 Social Media Data

- **Reddit:** r/SuicideWatch, r/depression subreddits
- **Twitter/X:** Suicide-related hashtags and keywords
- **Facebook:** Public posts and groups

3.2 News Articles

- Major news outlets (Prothom Alo, The Daily Star)
- Regional and local newspapers
- Wire services (AP, Reuters)

3.3 Clinical Data

- Electronic health records (EHR)
- Clinical notes and discharge summaries
- Crisis hotline transcripts

3.4 Cross-Cultural Considerations

Table 2: Multilingual Datasets for Suicide Detection

Language	Dataset	Size	Reference
English	CLPsych	1,200 posts	Coppersmith et al. (2015)
Spanish	Spanish CLPsych	800 posts	Garcia et al. (2021)
Bangla	Prothom Alo	10,000+ articles	This work
Hindi	Hindi CLPsych	500 posts	Sharma et al. (2022)

4 Feature Engineering

4.1 Text-Based Features

- **Lexical:** Word frequency, character n-grams, vocabulary richness
- **Syntactic:** POS tags, dependency patterns, sentence complexity
- **Semantic:** Word embeddings, topic distributions, named entities

4.2 Psycholinguistic Features

- LIWC (Linguistic Inquiry and Word Count) categories
- Emotion lexicons (NRC, ANEW)
- Suicide-specific keyword dictionaries

4.3 Cross-Cultural Features

- Regional triggers (farm distress, dowry, academic pressure)
- Demographic indicators
- Cultural context markers

5 Clustering and Pattern Analysis

5.1 Clustering Algorithms

- **K-Means:** Partitional clustering for distinct risk groups
- **Agglomerative:** Hierarchical clustering for taxonomic analysis
- **DBSCAN:** Density-based clustering for outlier detection
- **HDBSCAN:** Hierarchical density-based clustering

5.2 Topic Modeling

- **LDA:** Latent Dirichlet Allocation for thematic discovery
- **BERTopic:** Neural topic modeling with BERT embeddings
- **LLM-based:** Topic extraction using large language models

5.3 Cluster Validation

- Internal metrics: Silhouette score, Calinski-Harabasz index
- External metrics: Normalized Mutual Information, Adjusted Rand Index
- Custom metrics for balanced cluster assessment

6 Risk Scoring and Classification

6.1 Risk Level Classification

- Binary: High risk vs. low risk
- Multi-class: No risk, low, moderate, high, imminent
- Continuous: Risk score (0–1 scale)

6.2 Regional Trigger Categorization

Table 3: Regional Trigger Categories

Trigger Category	Description
Farm Distress	Agricultural failure, crop loss
Dowry	Marriage-related financial pressure
Academic	Educational stress, exam pressure
Economic	Unemployment, poverty
Family Conflict	Domestic violence, marital issues

6.3 Validation Methods

- Cross-validation strategies
- Temporal validation for time-series data
- Stratified sampling for imbalanced datasets

7 Ethical Considerations

7.1 Privacy and Anonymization

- De-identification of personal information
- Case ID hashing instead of names
- Geographic generalization (district centroids)

7.2 Responsible Reporting

- Avoiding sensationalism in reporting
- Including help-seeking resources
- Following media guidelines (WHO recommendations)

7.3 Data Sensitivity

- Handling vulnerable populations
- Informed consent considerations
- Institutional review board (IRB) requirements

8 Research Gaps

8.1 Current Limitations

1. Limited datasets for South Asian languages
2. Lack of culturally-aware models
3. Insufficient cross-cultural validation
4. Privacy concerns limiting data sharing
5. Generalizability across regions

8.2 Future Directions

1. Multilingual model development for under-represented languages
2. Cross-cultural transfer learning
3. Real-time monitoring systems
4. Integration with public health interventions
5. Explainable AI for clinical deployment

9 Conclusion

This literature review has surveyed the current state of suicide risk detection using NLP and machine learning methods. While significant progress has been made in English-language contexts, there remains a critical need for research in under-represented languages and cultures, particularly in South Asia. The integration of large language models with culturally-aware feature engineering presents promising opportunities for developing more effective and equitable suicide prevention systems.

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